An NYC Airbnb Market Intelligence Platform

An End-to-End Data Warehouse & AI Implementation on Snowflake

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From Raw Data to Actionable Strategy

The Challenge

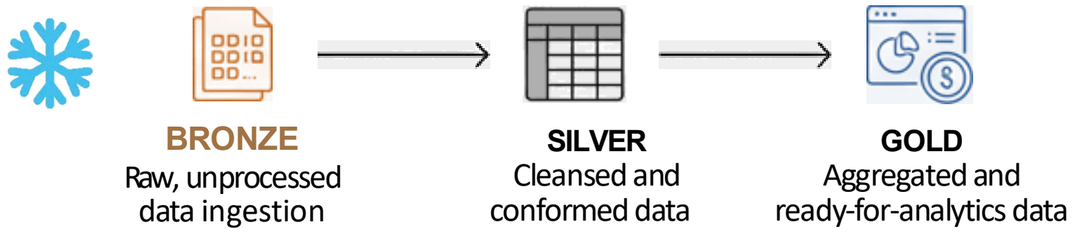
Ingest and model a large, complex Airbnb dataset for New York City

Build a robust, scalable, and automated data platform on cloud

Support both business intelligence and advanced AI-driven analytics

Our Solution: The Medallion Architecture

- An end-to-end data pipeline built entirely on the Snowflake Data Cloud
- A Medallion Architecture (Bronze, Silver, Gold) to progressively refine raw listings into strategic insights
- Augmented with Snowflake Cortex AI to unlock deeper intelligence from both structured and unstructured data



Stage 1: Ingesting Raw Data into the Bronze Layer

Objective:

To load the complete, unaltered Airbnb dataset, creating a persistent, queryable staging area for all downstream processing

Key Actions:

- Loaded all source files into a single Bronze table (BRONZE.RAW_AIRBNB_LISTINGS), establishing a centralized and reliable raw-data repository.
- Captured operational metadata (ingestion timestamp, source filename, record hash) to ensure full data lineage and auditability.
- Preserved original fields and data fidelity, enabling flexible downstream transformations, analytics, and ML feature engineering.

```sql
create or replace TABLE DB_TEAM_SAR.BRONZE.RAW_AIRBNB_LISTINGS (
    ID NUMBER(38,0) NOT NULL,
    NAME VARCHAR(500),
    HOST_ID NUMBER(38,0),
    HOST_IDENTITY_VERIFIED VARCHAR(20),
    HOST_NAME VARCHAR(200),
    NEIGHBOURHOOD_GROUP VARCHAR(100),
    NEIGHBOURHOOD VARCHAR(100),
    LAT FLOAT,
    LONG FLOAT,
    COUNTRY VARCHAR(100),
    COUNTRY_CODE VARCHAR(10),
    INSTANT_BOOKABLE BOOLEAN,
    CANCELLATION_POLICY VARCHAR(50),
    ROOM_TYPE VARCHAR(100),
    CONSTRUCTION_YEAR NUMBER(38,0),
    PRICE NUMBER(38,0),
    SERVICE_FEE NUMBER(38,0),
    MINIMUM_NIGHTS NUMBER(38,0),
    NUMBER_OF_REVIEWS NUMBER(38,0),
    LAST_REVIEW DATE,
    REVIEWS_PER_MONTH FLOAT,
    REVIEW_RATE_NUMBER NUMBER(38,0),
    CALCULATED_HOST_LISTINGS_COUNT NUMBER(38,0),
    AVAILABILITY_365 NUMBER(38,0),
    HOUSE_RULES VARCHAR(16777216),
    INGESTION_TIMESTAMP TIMESTAMP_NTZ(9) DEFAULT CURRENT_TIMESTAMP(),
    SOURCE_FILE VARCHAR(255),
    RECORD_HASH VARCHAR(64),
    IS_CURRENT BOOLEAN DEFAULT TRUE,
    BATCH_ID VARCHAR(50),
    HOUSE_RULES_SENTIMENT VARCHAR(20) COMMENT 'AI: Positive/Negative/Neutral',
    HOUSE_RULES_SUMMARY VARCHAR(1000) COMMENT 'AI: Generated summary',
    PRICE_CATEGORY VARCHAR(50) COMMENT 'AI: Price classification',
    LISTING_QUALITY_SCORE FLOAT COMMENT 'ML: Predicted quality score',
    LAST_UPDATED TIMESTAMP_NTZ(9),
    primary key (ID)
)COMMENT='Raw Airbnb listing data from source system'
;
```

	ID	NAME	HOST_ID	HOST_IDENTITY_VERIFIED	HOST_NAME	NEIGHBOURHOOD_GROUP	NEIGHBOURHOOD
1	1001254	Clean & quiet apt home by the park	80014485718	unconfirmed	Madaline	Brooklyn	Kensington
2	1002102	Skylit Midtown Castle	52335172823	verified	Jenna	Manhattan	Midtown
3	1003689	Entire Apt: Spacious Studio/Loft by central park	92037596077	verified	Lyndon	Manhattan	East Harlem
4	1004098	Large Cozy 1 BR Apartment In Midtown East	45498551794	verified	Michelle	Manhattan	Murray Hill
5	1005202	BlissArtsSpace!	90821839709	unconfirmed	Emma	Brooklyn	Bedford-Stuyvesant

Stage 2: Build Silver layer using Star Schema

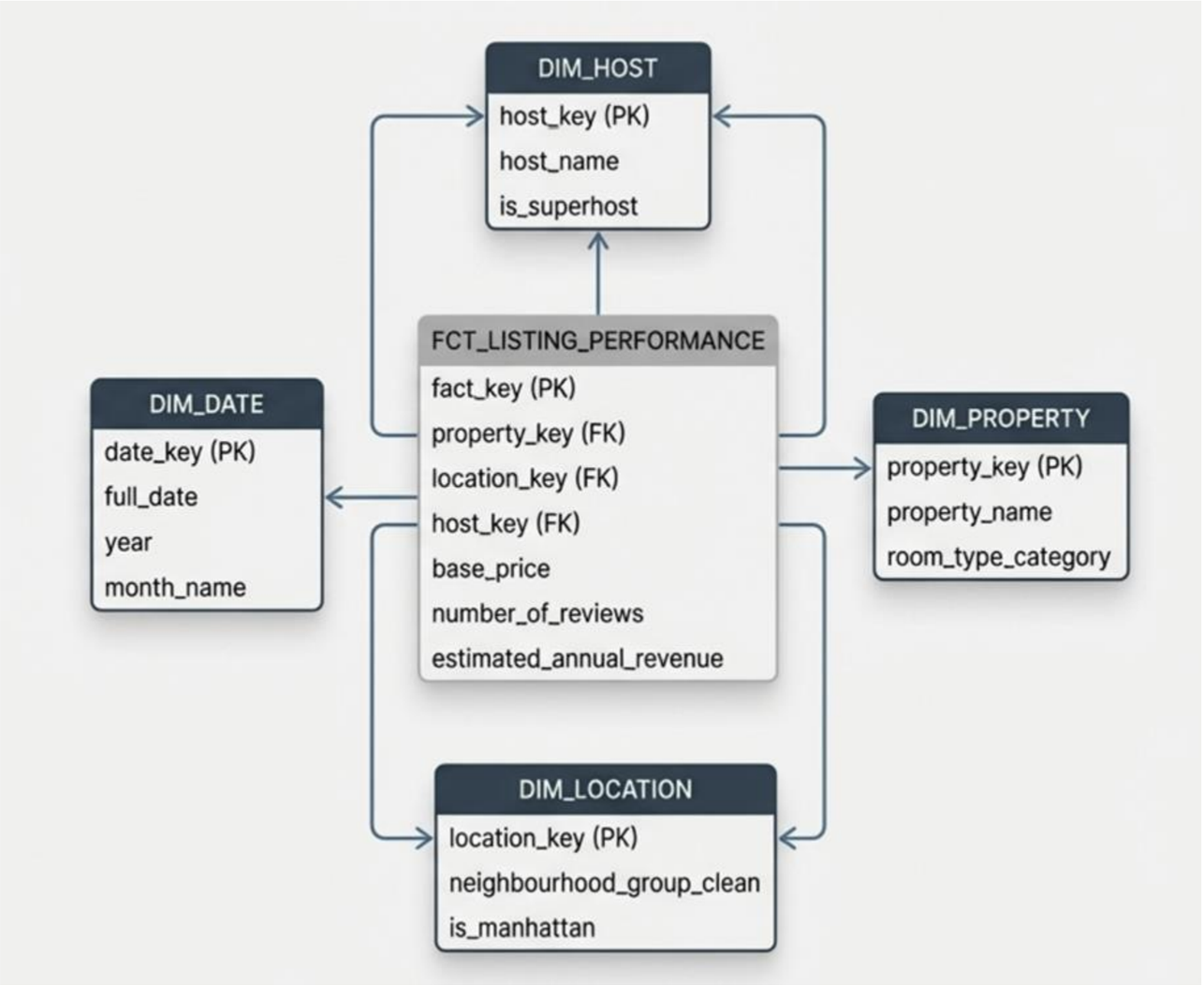
Objective:

To transform raw data into a clean, integrated, and high-performance model for enterprise-wide analysis

Key Features:

- Fact Table - FCT_LISTING_PERFORMANCE
  Centralizes key performance metrics like price, number_of_reviews, and calculated fields such as estimated annual_revenue

- Dimension Tables
  DIM_HOST
  DIM_LOCATION
  DIM_DATE
  DIM_PORPERTY



Stage 2: Build Silver layer using Star Schema

Enforced relationships between fact and dimension tables ensure data consistency and high-performance joins.

```sql
create or replace view DB_TEAM_SAR.SILVER.VW_AIRBNB_SEMANTIC(
    LISTING_ID,
    BASE_PRICE,
    NUMBER_OF_REVIEWS,
    ESTIMATED_MONTHLY_REVENUE,
    ESTIMATED_ANNUAL_REVENUE,
    ESTIMATED_OCCUPANCY_RATE,
    REVIEW_SCORE_NORMALIZED,
    BOROUGH,
    NEIGHBORHOOD,
    ROOM_TYPE_CATEGORY,
    INSTANT_BOOKABLE,
    CANCELLATION_POLICY,
    IS_SUPERHOST,
    IDENTITY_VERIFIED
) as
```

Defines Airbnb semantic view with a curated, business-friendly column list

```sql
SELECT
    f.listing_id,
    f.base_price,
    f.number_of_reviews,
    f.estimated_monthly_revenue,
    f.estimated_annual_revenue,
    f.estimated_occupancy_rate,
    f.review_score_normalized,
    l.neighbourhood_group_clean as borough,
    l.neighbourhood_clean as neighborhood,
    p.room_type_category,
    p.instant_bookable,
    p.cancellation_policy,
    h.is_superhost,
    h.identity_verified
FROM FCT_LISTING_PERFORMANCE f
LEFT JOIN DIM_LOCATION l ON f.location_key = l.location_key
LEFT JOIN DIM_PROPERTY p ON f.property_key = p.property_key
LEFT JOIN DIM_HOST h ON f.host_key = h.host_key;
```

Joins Location, Property, and Host dimensions via surrogate keys to enrich the fact data with borough, room type, and host trust signals

Stage 3: Designing and loading data into Gold Layer

```sql
create or replace view DB_TEAM_SAR.GOLD.V_PRICE_CATEGORY_PERFORMANCE(
    PRICE_CATEGORY,
    LISTING_COUNT,
    AVG_PRICE,
    TOTAL_ANNUAL_REVENUE
) as
SELECT
    m.price_category,
    COUNT(*) as listing_count,
    ROUND(AVG(m.price), 2) as avg_price,
    ROUND(SUM(f.estimated_annual_revenue), 2) as total_annual_revenue
FROM Bronze.ML_FEATURES m
JOIN Silver.FCT_LISTING_PERFORMANCE f ON m.listing_id = f.listing_id
GROUP BY m.price_category
ORDER BY total_annual_revenue DESC;
```

Aggregates performance metrics (listing count, average price, and total revenue) for each AI-generated price category to understand revenue contribution by segment

Compares Superhosts and regular hosts across occupancy, revenue, pricing, reviews, and calculates the price premium Superhosts command.

```sql
SELECT
    h.is_superhost,
    COUNT(DISTINCT f.listing_id) as listing_count,
    AVG(f.base_price) as avg_price,
    AVG(f.estimated_occupancy_rate) * 100 as avg_occupancy_pct,
    AVG(f.estimated_monthly_revenue) as avg_monthly_revenue,
    AVG(f.review_score_normalized) as avg_review_score,

    -- Premium calculation
    AVG(f.base_price) - AVG(AVG(f.base_price)) OVER () as price_premium,
    (AVG(f.base_price) / NULLIF(AVG(AVG(f.base_price)) OVER (), 0) - 1) * 100 as price_premium_pct
FROM Silver.FCT_LISTING_PERFORMANCE f
JOIN Silver.DIM_HOST h ON f.host_key = h.host_key
GROUP BY h.is_superhost;
```

Stage 4: Answering Key Questions from Gold Layer

2. Pricing Strategy Optimization:

How do room type, cancellation policy, and property age affect revenue and occupancy?
-> by AGG PRICING STRATEGY

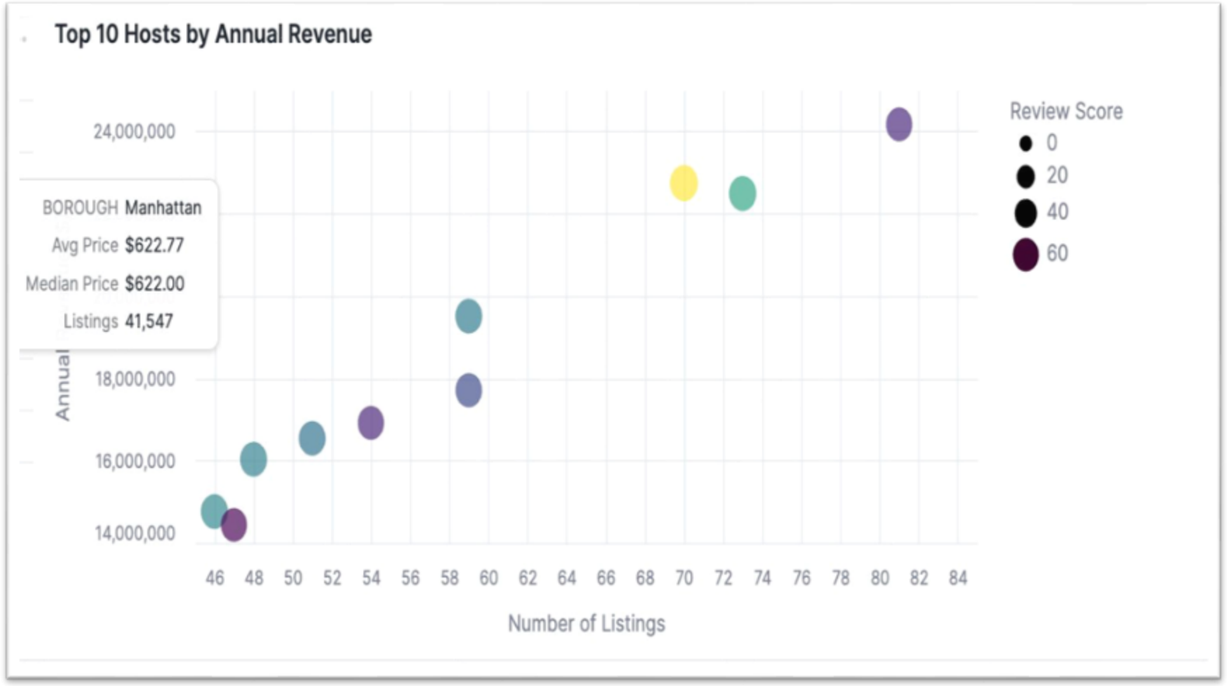
1. Location Performance Analysis

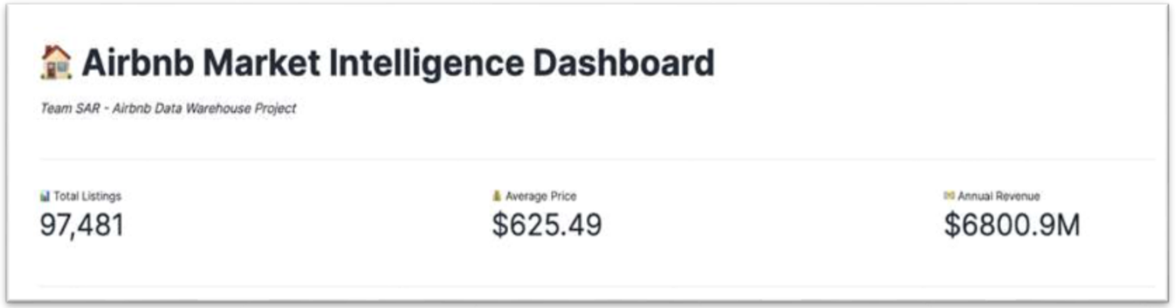
Which neighborhoods are most profitable? What is the market share and pricing power of each borough?
-> by AGG LOCATION PERFORMANCE

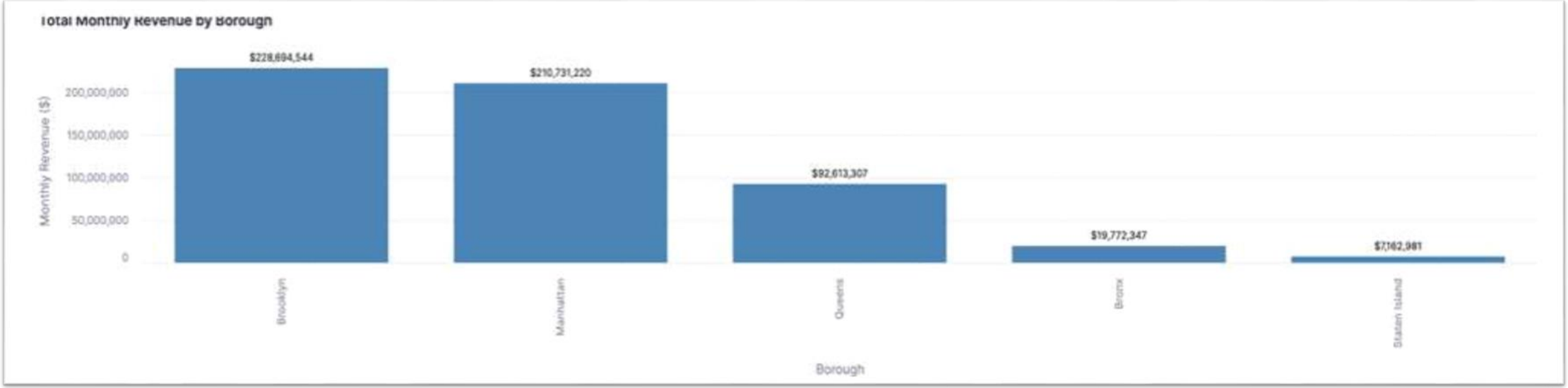
3. Host Performance Ranking

Who are the top- performing hosts by revenue, review score, and portfolio size?
-> by AGG HOST PERFORMANCE

Stage 4:
Streamlit Dashboard







Stage 5: Incremental Data Loading

Objective:

To ensure our platform stays current by loading new and updated listings without full reloads and refresh each of the layers

Process:

- Simulated a file of 100 new listings and loaded them into a staging table (‘STAGE1_NEW_LISTINGS ')
- Used a single ‘MERGE‘ statement to efficiently upsert these records into the BRONZE.INCREMENTAL_LOAD_LOG table
- Changes propagated to downstream Silver and Gold layers via our data transformation pipelines

```sql
MERGE INTO RAW_AIRBNB_LISTINGS target
USING STAGE_NEW_LISTINGS source
ON target.id = source.id
WHEN MATCHED THEN
    UPDATE SET
        target.price = source.price,
        target.last_updated = source.last_updated
WHEN NOT MATCHED THEN
    INSERT (
        id, NAME, host_id, host_identity_verified, host_name,
        neighbourhood_group, neighbourhood, lat, long, country, country_code,
        instant_bookable, cancellation_policy, room_type, Construction_year,
        price, service_fee, minimum_nights, number_of_reviews, last_review,
        reviews_per_month, review_rate_number, calculated_host_listings_count,
        availability_365, house_rules, ingestion_timestamp, source_file,
        record_hash, is_current, batch_id, house_rules_sentiment,
        house_rules_summary, price_category, listing_quality_score, last_updated
    )
    VALUES (
        source.id, source.NAME, source.host_id, source.host_identity_verified,
        source.host_name, source.neighbourhood_group, source.neighbourhood,
        source.lat, source.long, source.country, source.country_code,
        source.instant_bookable, source.cancellation_policy, source.room_type,
        source.Construction_year, source.price, source.service_fee,
        source.minimum_nights, source.number_of_reviews, source.last_review,
        source.reviews_per_month, source.review_rate_number,
        source.calculated_host_listings_count, source.availability_365,
        source.house_rules, source.ingestion_timestamp, source.source_file,
        source.record_hash, source.is_current, source.batch_id,
        source.house_rules_sentiment, source.house_rules_summary,
        source.price_category, source.listing_quality_score, source.last_updated
    );
```

Stage 5: Incremental Data Loading

Before load

Total Records

97994

+100

After load

Total Records

98094

```sql
SELECT
    'Original Data' as dataset,
    (SELECT records_loaded FROM INCREMENTAL_LOAD_LOG
     WHERE load_type = 'INITIAL_LOAD' LIMIT 1) as count
UNION ALL
SELECT
    'After Incremental Load',
    COUNT(*)
FROM RAW_AIRBNB_LISTINGS;
```

Results (just now)

Table Chart 2 rows 413ms

	DATASET	COUNT
1	Original Data	97994
2	After Incremental Load	98094

Stage 6: AI/ML functions on Bronze Layer

Objective:

Use Snowflake Cortex AI functions to analyze unstructured text in the 'house rules column and classify structured data, adding new analytical dimensions to our raw data.

HOUSE_RULES	PRICE	HOUSE_RULES_SENTIMENT	HOUSE_RULES_SUMMARY	PRICE_CATEGORY
Guests are required to remove shoes upon entry. Quiet hours are strictly enforced from 10 PM to 7 AM. No parties or events allowed. Please keep the kitchen clean after use. Pets are not permitted in the bedrooms. Smoking is strictly prohibited anywhere on the property.	$450	Neutral	Strict rules include no shoes, quiet hours 10 PM-7 AM, no parties, kitchen cleanlIness, no pets in bedrooms, and no smoking.	Premium
Feel free to use the kitchen and living room. Please be respectful of the neighbors. Enjoy your stay!	$120	Positive	Use kitchen/living room, respect neighbors, enjoy stay.	Budget
No loud noise after 9 PM. Trash must be taken out on Tuesday nights.	$275	Neutral	Quiet after 9 PM; trash out on Tuesdays.	Mid-Range

Key Functions Deployed:

Sentiment Analysis: SNOWFLAKE.CORTEX.SENTIMENT

Analyzed ”house_rules” text to determine if the host's tone was Positive, Neutral, or Negative.

Summarization: SNOWFLAKE.CORTEX.SUMMARIZE

Generated concise summaries of lengthy 'house_rules‘ descriptions

Classification: SNOWFLAKE.CORTEX.COMPLETE’):
Classified listing ‘price’ into business-friendly categories (Budget, Mid-Range, Premium, Luxury) using a Cortex LLM

Stage 7: Cortex Search

Objective:

To enable fast, intuitive, semantic search over our entire property catalog, allowing users to find listings without needing to know specific keywords or filters.

Implementation:

- Created a CORTEX SEARCH SERVICE‘ named AIRBNB PROPERTY SEARCH‘ on the RAW_AIRBNB_LISTINGS‘ table.
- Indexed key fields including name , house rules, and neighbourhood to power the search.

```sql
CREATE CORTEX SEARCH SERVICE AIRBNB_PROPERTY_SEARCH
ON name
WAREHOUSE = ANIMAL_TASK_WH
TARGET_LAG = '1 hour'
AS (
    SELECT
        id,
        name,
        house_rules,
        room_type,
        neighbourhood_group,
        instant_bookable,
        cancellation_policy,
        price,
        review_rate_number,
        availability_365
    FROM RAW_AIRBNB_LISTINGS
    WHERE name IS NOT NULL
);
```

Stage 7: Cortex Search

Search 1

Pet-friendly apartments

Table | Chart — 10 rows — 388ms

	ID	NAME	NEIGHBOURHOOD	ROOM_TYPE	PRICE	RATING
	52789...55146...	Pet-friendly 1bd in the Heart of East Village 20.0% A spacious sunny room in NYC PET Friendly 10.0% +7 more	Manhattan 80.0% Queens 10.0% +1 more	Entire home/a... 70.0% Private room 30.0%	78 763	2 5
1	5278900	Pet friendly, homey 1 BR in Lower East Side, NYC	Manhattan	Entire home/apt	763	5
2	37935846	Modern 1BR in Upper West Side, Pet-Friendly	Manhattan	Entire home/apt	620	3
3	36619715	Pet-Friendly UES One Bedroom	Manhattan	Entire home/apt	514	5
4	20388171	Pet-friendly 1bd in the Heart of East Village	Manhattan	Entire home/apt	120	4
5	55146067	Pet-friendly 1bd in the Heart of East Village	Manhattan	Entire home/apt	120	4
6	32996075	Pet-friendly Central Park Loft w/ Private Terrace	Manhattan	Private room	581	3
7	35535550	Stylish 1BR in Upper West Side, Pet-Friendly	Manhattan	Entire home/apt	585	2
8	29903195	A spacious sunny room in NYC PET Friendly	Manhattan	Private room	236	3
9	29871714	Cozy pet-friendly one bedroom apartment.	Staten Island	Entire home/apt	78	2
10	52323814	LIC 1 Bedroom in a pet-friendly apartment.	Queens	Private room	524	5

Search 2

Luxury spacious properties

	ID	NAME	NEIGHBOURHOOD	PRICE	RATING
	11510500 53940948	Beautiful and Spacious Apartment 20.0% Luxury spacious new 1Bdr 30 mins to Manhatt... 20.0% +6 more	Brooklyn 40.0% Manhattan 40.0% +1 more	135 1191	3 5
1	30360499	Million dollar view Luxury + spacious apartment	Manhattan	1191	5
2	28551165	Million dollar view in luxury spacious apartment	Manhattan	1015	3
3	53940948	Luxury spacious new 1Bdr 30 mins to Manhattan	Brooklyn	626	5
4	19183052	Luxury spacious new 1Bdr 30 mins to Manhattan	Brooklyn	626	5
5	53381468	Beautiful and Spacious Apartment	Queens	591	5
6	18623572	Beautiful and Spacious Apartment	Queens	591	5
7	26472307	Modern Spacious Apartment	Brooklyn	281	5
8	11510500	Spacious Room In Luxury NYC Apartment	Manhattan	273	3
9	29789974	Luxury, Furnished, Spacious Apartment - 2BR / 2BA	Brooklyn	158	4
10	50661391	Spacious apartment in Manhattan	Manhattan	135	4

Search 3

Free wifi

	ID	NAME	NEIGHBOURHOOD	ROOM_TYPE	PRICE	RATING
	1176415 391398...	One Bedroom Mini studio - Free... 30.0% 2 Beds! Free wifi/cable. Very cl... 10.0% +6 more	Queens 50.0% Brooklyn 30.0% +2 more	Entire home/apt 90.0% Private room 10.0%	267 1081	1 5
1	31238104	Convenient , modern w/ free WiFi.	Brooklyn	Entire home/apt	781	3
2	35453810	5 G free WiFi in room.	Manhattan	Private room	485	5
3	22224568	Staten Island - Free Wifi, Parking Space, Ne	Staten Island	Entire home/apt	436	4
4	2176415	One Bedroom Mini studio - Free WIFI (New)	Queens	Entire home/apt	673	1
5	36238076	2 Beds! Free wifi/cable. Very close 2 LGA.	Queens	Entire home/apt	419	2
6	4161597	PRIVATE 1BR APT- Free WIFI & DIRECT TV	Brooklyn	Entire home/apt	267	1
7	36940049	Private Home Away From Home FREE WIFI-	Brooklyn	Entire home/apt	628	5
8	32803875	One Bedroom Mini studio - Free WIFI	Queens	Entire home/apt	815	3
9	1176415	One Bedroom Mini studio - Free WIFI	Queens	Entire home/apt	612	1
10	39139860	One Bedroom Mini studio - Free WIFI	Queens	Entire home/apt	1081	4

Stage 8: Cortex Analyst

Objective:
To empower business users to query our structured Silver Layer data using natural language, without writing a single line of SQL

Q. What is the average listing price?

AIRBNB_SEMANTIC_MODEL

User
what is the average listing price ?

Cortex Analyst Request ID
This is our interpretation of your question:

What is the average listing price?

AVERAGE_LISTING_PRICE
625.487652

Logical query | Physical query

```
SELECT
  AVG(base_price) AS average_listing_price
FROM
  vw_airbnb_semantic
WHERE
  NOT base_price IS NULL
  /* Generated by Cortex Analyst */
```

Q. Show me total revenue by borough

User
show me total revenue by borough

Cortex Analyst Request ID
This is our interpretation of your question:

Show me total revenue by borough over the entire available time period. I'll use estimated_annual_revenue as the revenue measure since it represents the total income generated by listings.

BOROUGH	TOTAL_REVENUE
Brooklyn	2782450288.74
Manhattan	2563896506.715
Queens	1126795229.205
Bronx	240563552.125
Staten Island	87149604.235

Logical query | Physical query

```
SELECT
  borough,
  SUM(estimated_annual_revenue) AS total_revenue
FROM
  vw_airbnb_semantic
WHERE
  NOT borough IS NULL
GROUP BY
  borough
ORDER BY
  total_revenue DESC NULLS LAST
  /* Generated by Cortex Analyst */
```

Q. Do super hosts earn more revenue than regular hosts?

User
do superhosts earn more revenue than regular hosts?

Cortex Analyst Request ID
This is our interpretation of your question:

Compare the average estimated annual revenue between superhosts (is_superhost = TRUE) and regular hosts (is_superhost = FALSE) to determine if superhosts earn more revenue than regular hosts

IS_SUPERHOST	HOST_COUNT	AVG_ANNUAL_REVENUE	TOTAL_AN
TRUE	5111	123412.28965271	630760
FALSE	92883	66428.678752893	617009

Logical query | Physical query

```
SELECT
  is_superhost,
  COUNT(*) AS host_count,
  AVG(estimated_annual_revenue) AS avg_annual_revenue,
  SUM(estimated_annual_revenue) AS total_annual_revenue
FROM
  vw_airbnb_semantic
WHERE
  NOT is_superhost IS NULL
GROUP BY
  is_superhost
ORDER BY
  avg_annual_revenue DESC NULLS LAST
  /* Generated by Cortex Analyst */
```

Project Summary & Key Achievements

We successfully designed and deployed a full-scale market intelligence platform for NYC Airbnb data, transforming raw files into a dynamic, AI-powered analytical asset on Snowflake.

End-to-End Pipeline

Implemented a robust Bronze, Silver, and Gold Medallion architecture for reliable data processing.

Actionable Insights

Developed a star schema and aggregated data marts to answer key business questions, visualized in a Streamlit dashboard.

AI Integration

Leveraged Snowflake Cortex for sentiment analysis, summarization, summarization, semantic search, and natural language querying.

Dynamic & Scalable

Engineered the system to handle incremental data loads, ensuring it remains current and reliable for ongoing analysis.

THANK YOU